SIMEGNEW AREGAHEGN

Data Science Student | AI/ML | Computer Vision | Cloud & DevOps

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Professional Summary

Data Science student at Debre Birhan University with hands-on experience in machine learning, deep learning, natural language processing, computer vision, data analysis, and software development. Experienced in developing AI projects involving Amharic–English machine translation, environmental prediction, and web-based AI applications. Familiar with Python, PyTorch, TensorFlow, scikit-learn, Git/GitHub, React, and data-processing tools. Currently developing skills in cloud computing and DevOps and seeking an internship opportunity to apply technical knowledge to real-world projects while continuing to grow professionally.

Education

Bachelor's Degree in Data Science **2022 – Present**

Debre Birhan University, Ethiopia

Technical Skills

Programming: Python, JavaScript, HTML, CSS, SQL, R

Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Scikit-learn

Deep Learning: PyTorch, TensorFlow, Keras

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Analysis

NLP: SentencePiece, Neural Machine Translation, Text Preprocessing

Web Development: React, Vite, HTML, CSS, JavaScript

Big Data: Apache Spark, ETL, Data Processing

Cloud & DevOps: Cloud Computing, DevOps Fundamentals

Tools: Git, GitHub, Google Colab, Kaggle, Jupyter Notebook

Selected Projects

Amharic ↔ English Neural Machine Translation

- Developed a research project focused on bidirectional Amharic–English machine translation.
- Built a preprocessing pipeline for cleaning, normalization, language filtering, deduplication, and parallel-corpus preparation.
- Prepared and processed a large parallel corpus containing more than 200,000 sentence pairs.
- Explored Transformer-based neural machine translation and SentencePiece tokenization.
- Worked with PyTorch and GPU-based training environments.
- Explored evaluation using BLEU and chrF metrics.

Soil Erosion Prediction Using Deep Learning

- Developed a deep learning project for predicting soil erosion using environmental and geographical features.
- Worked with features including elevation, rainfall, slope, NDVI, land use, soil type, drainage density, and terrain variables.
- Investigated data quality and potential data leakage issues.
- Applied machine learning and deep learning techniques to an environmental prediction problem.

AI Coding Assistant Website

- Designed an AI-powered coding assistant web application.
- Developed frontend components using React and modern web technologies.
- Explored AI service integration, APIs, authentication, and database-backed application architecture.
- Applied software development and web application design principles.

Personal Portfolio Website

- Developed a personal portfolio website to showcase technical skills and projects.
- Built using React, HTML, CSS, and JavaScript.
- Organized projects, technical skills, services, and contact information into a professional portfolio.
- Deployed the portfolio online.

Practical Experience

AI Internship Project **2026**

Ethiopian Artificial Intelligence Institute, Ethiopia

- Participated in an AI-focused internship/project environment.
- Collaborated with team members on an AI project under supervision.
- Applied machine learning and data-processing concepts to practical project work.
- Participated in technical discussions, project planning, and implementation.

Additional Learning

- Cloud Computing and DevOps
- Computer Vision
- Natural Language Processing
- Deep Learning
- Big Data and Apache Spark
- Machine Learning
- Data Engineering Concepts
- Git and GitHub

Languages

Amharic: Native **English:** Working Proficiency

References

Available upon request.